Advanced Programming Practice · 21CSC203P

Unit 4 · Session 3 · SLO 1 — Solved Worksheet

Name: AKARSH MURALI AKARSH MURALI | **Reg. No:** RA2511030010067

Unit – 4 : Session – 3 : SLO – 1 | Custom Higher-Order Functions

I. Fill in the Blanks

1. lambda — a lambda function in Python is a short anonymous function.
2. function — a higher-order function can take another function as an argument (or return one).

II. Multiple Choice

1. list(filter(lambda x: x % 2 == 0, [1, 2, 3, 4])) → [2, 4] — filter keeps only the even numbers.

III. Example — Custom Higher-Order Function

• def apply_twice(func, value): return func(func(value))

• apply_twice(lambda x: x * 3, 2) # → 18

• Higher-order functions treat functions as first-class values: they can be passed, returned and stored like any other object.